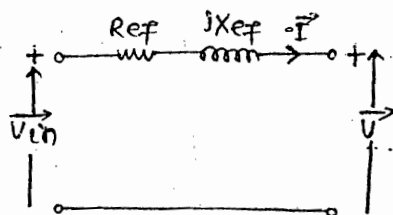


VOLTAGE REGULATION

- * VR of the Xmer is defined as the rise in o/p vol. expressed as a fraction of FL rated vol. when FL at a specified PF is reduced to 0, keeping the i/p vol. constant.



$$VR^{\%} = \frac{V_{in} - V}{V} \text{ pu}$$

Note:- Use magnitude only

where, $V = V_{rated}$

$$\vec{V}_{in} = \vec{V} + \vec{I} \vec{Z}_{eq}$$

$$VR^{\%} = \left(\frac{V_{in}}{V} - 1 \right) \text{ pu} = (V_{in})_{\text{pu}} - 1 \text{ pu}$$

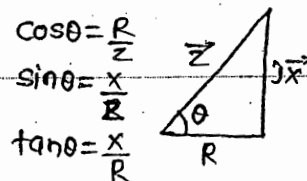
Que. → The TF has pu imp. of 0.10 pu & pu resistance of 0.01. Calc. the phase diff. betⁿ the o/p Vol. & i/p Vol. of FL on
(a) 0.8 PF lag (b) Unity PF (c) 0.8 PF lead

And hence determine the VRⁿ under the above condⁿ.

Solⁿ →

$$Z_{eq} = 0.10 \angle \cos^{-1} \frac{0.01}{0.10}$$

$$Z_{eq} = 0.01 + j0.0994 \text{ pu} \\ = 0.1 \angle 84.26^{\circ} \text{ pu}$$



$$(i) V_{in} = 1 \angle 0^{\circ} + 1.0 \angle -\cos^{-1} 0.8 \times 0.1 \angle 84.26^{\circ}$$

$$= 1.0702 \angle 3.94^{\circ} \text{ pu}$$

$$VR^{\%} = 1.0702 - 1 \text{ pu}$$

$$= 0.0702 \text{ pu}$$

$$\boxed{VR^{\%} = 7.02\%} \text{ (MORE)}$$

$$(ii) V_{in} = 1 \angle 0^{\circ} + 1.0 \angle \cos^{-1}(1) \times 0.10 \angle 84.26^{\circ}$$

$$= 1.015 \angle 5.63^{\circ} \text{ pu}$$

$$VR^{\%} = 1.015 - 1$$

$$= 0.015$$

$$\boxed{VR^{\%} = 1.5\%} \text{ (IMPROVES)}$$

$$(ii) V_{in} = 1 \angle 0^{\circ} + 1.0 \angle \cos^{-1} 0.8 \times 0.1 \angle 84.26^{\circ}$$

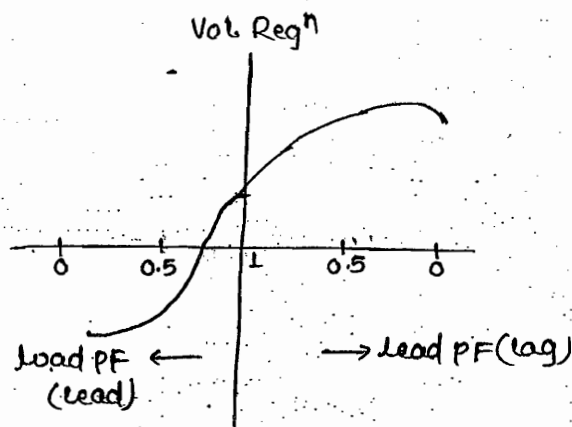
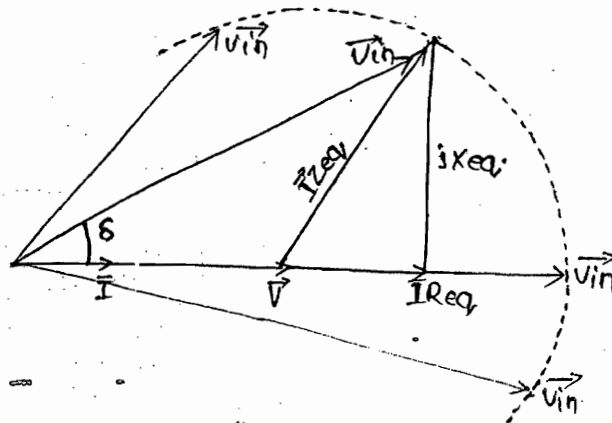
$$= 0.9522 \angle -1.157^{\circ}$$

$$VR^{\%} = 0.9522 - 1$$

$$VR^{\%} = -0.0478$$

$$\boxed{VR^{\%} = -4.78\%} \text{ (LESS)}$$

locus of V_{in} at diff PF



$$0.8 \text{ PF } \text{Regn} = V_{in} = 1.0702 / 3.94$$

$$V_{in} \cos \delta = 1.0702 \cos(3.94)$$

$$= 1.0675$$

$$V_{Rn} = 1.0675 - 1$$

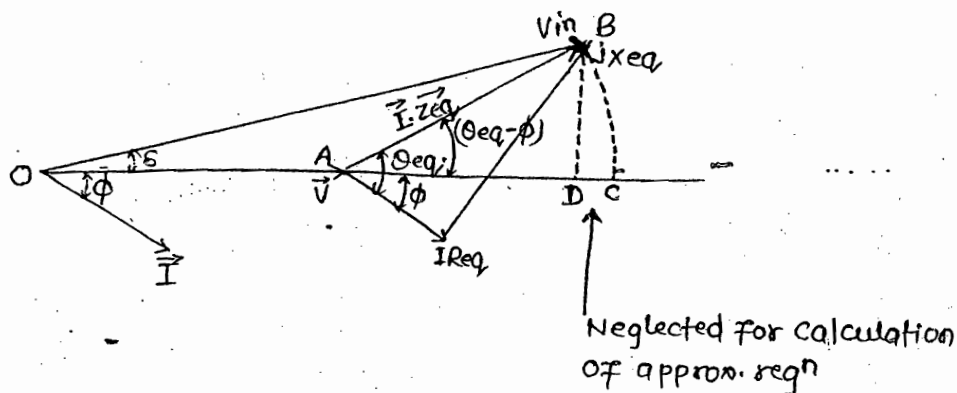
$$\% \text{VR} = 6.75\%$$

$$\% \text{ error} = \frac{7.02 - 6.75}{7.02}$$

$$= 3.44\%$$

→ previous value (0.8 PF lag)

* Approximate Regulation $\rightarrow$ (Graphical approach)



$$\begin{aligned} \text{Actual } VR^n &= \frac{OB - OA}{OA} \\ &= \frac{OC - OA}{OA} \end{aligned}$$

Since δ is very small, $V_{in} \cos \delta \approx V_{in}$
i.e. $OP \approx OC$

$$\text{Then approximate } v_R^n = \frac{OD - OA}{OA} = \frac{AD}{OA} = \frac{AB \cdot \cos(\theta_{eq} - \phi)}{OA}$$

$$\text{Approximate } VR^n = \frac{I \cdot Z_{eq} \cos(\theta_{eq} - \phi)}{V}$$

$$= Z_{pu} \cos(\theta_{eq} - \phi)$$

where ϕ is the tve for lagging pf.

Analytical approach →

$$V_{in} \angle \delta = V \angle 0 + I \angle -\phi \times Z_{eq} \angle \theta_{eq}$$

$$= V + I Z_{eq} (\theta_{eq} - \phi)$$

$$V_m \cos \delta + j V_m \sin \delta = V + [I_{eq} Z_{eq} \cos(\theta_{eq} - \phi)] + j [I_{eq} Z_{eq} \sin(\theta_{eq} - \phi)]$$

Equating Real parts

$$V_{in} \cos \delta = V + I Z_{eq} \cos(\theta_{eq} - \phi)$$

Since δ is very small

$\sin \cos \sin$ $\sin$ for calⁿ of approx Regⁿ

Then; $V_{in} = V + I Z_{eq} \cos(\theta_{eq} - \phi)$

$V_{in} - V = I Z_{eq} \cos(\theta_{eq} - \phi)$

approx Regn = $\frac{V_{in} - V}{V}$
 $= \frac{I Z_{eq} \cos(\theta_{eq} - \phi)}{V}$

i.e. approx Regn = $Z_{pu} \cos(\theta_{eq} - \phi)$ (where ϕ is +ve for lagging PF)

* Maximum Regulation $\rightarrow$

Where $\theta_{eq} - \phi = 0^\circ$

$\phi = \theta_{eq}$

$\therefore \text{max}^m \text{Regn} = Z_{pu}$

$\therefore$ Corresponding PF = $\cos \phi$

= $\cos \theta_{eq}$ (lagging)

= $\frac{R_{ef}}{Z_{ef}}$ (lagging)

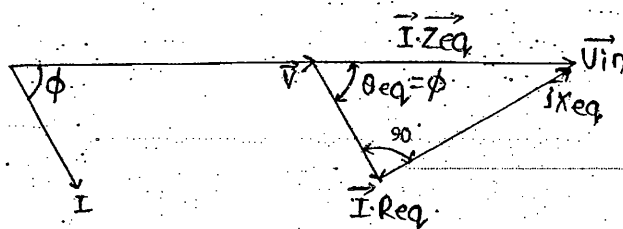


Fig:- Phasor dia. at max^m Regulation

(No approximation is involved $\because \delta = 0$)

* Zero Regn (approx condn) $\rightarrow$

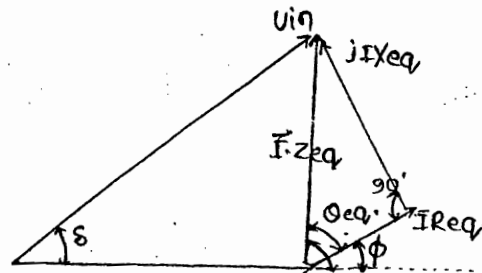
when $\theta_{eq} - \phi = 90^\circ$

$\phi = (\theta_{eq} - 90^\circ)$ lagging

= $-(90 - \theta_{eq})$ lagging

= $(90 - \theta_{eq})$ leading

$$\begin{aligned}
 \therefore \text{Corresponding PF} &= \cos(90^\circ - \theta_{eq}) \text{ leading} \\
 &= \sin \theta_{eq} \text{ (leading)} \\
 &= \frac{X_{eq}}{Z_{eq}} \text{ (leading)}
 \end{aligned}$$



* min^m Regulation (approximate condⁿ) →

$$\text{Approximate cond}^n = Z_{pu} \cos(\theta_{eq} - \phi)$$

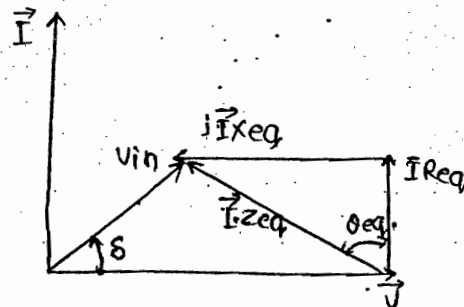
min Regⁿ at $\phi = 90^\circ$ leading i.e. at zero PF leading

$$\text{min}^n \text{Reg}^n = Z_{pu} \cdot \cos[\theta_{eq} - (-90^\circ)]$$

$$= Z_{pu} \cdot \cos(\theta_{eq} + 90^\circ)$$

$$= -Z_{pu} \sin \theta_{eq}$$

$$= -X_{pu}$$



* VR is a figure of merit of a TF & its low value is desired.

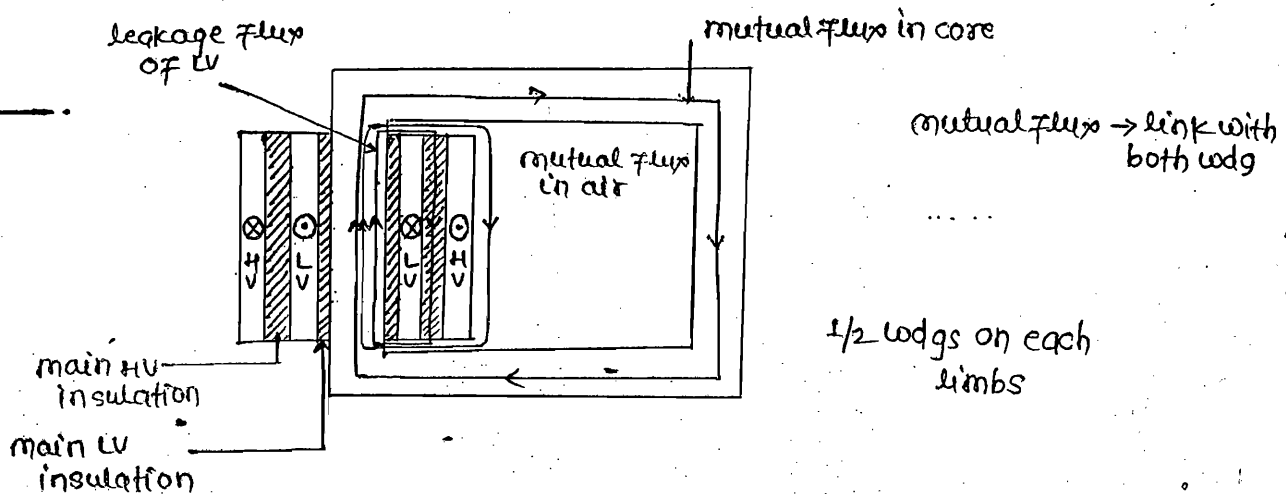
* VR may be reduced by reducing the pu impedance of TF.

* Z_{pu} can be reduced by reducing (R_{pu} &/or) X_{pu} .

* R_{pu} is already kept at an optimally low value, because of η consideration.

* This leaves X_{pu} to be reduced for reducing the value of pu impedance.

* X_{pu} can be reduced by reducing the leakage flux.



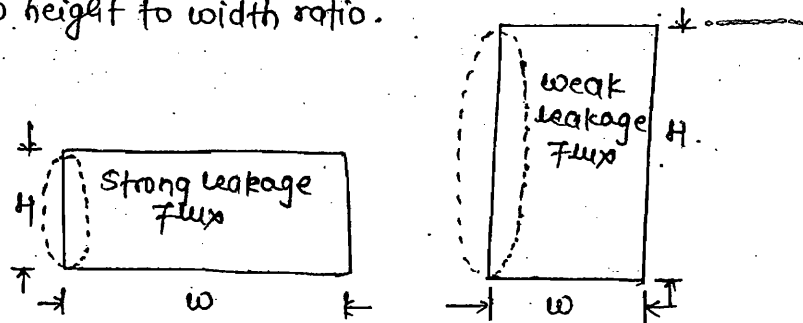
* Leakage flux can be reduced by keeping the wdg's physically close together.

This is obtain in core type Xmer by using concentric cylindrical wdg's.

In shell type TF the physical proximity bet'n wdg is obtained by using sandwich wdg's, also called pancake wdg's (or) interleaved wdg's.

In fact the leakage reactance of shell type TF may be graded as per requirement by adjusting the thickness of each layer &/or arranging the layers of HV & LV sections.

* In core type Xmer the leakage reactance may also be reduced by increasing the window height to width ratio.



Power TF ON $\rightarrow$ Always on FL

* A power TF operates on FL (or) is switched off.

Therefore UR^n as a performance index is not a significant factor in Power TF.

However the load on a distribution TF depends upon the consumers demand. And therefore may vary from FL to NL.

Obviously therefore UR^n is a very significant factor in distribution TF & should therefore kept low.

Accordingly the pu impedance of the distribution TF may be as low as 0.015 pu while for a power TF it may be as high as 0.15 pu.

A high value of pu impedance in power TF has the advantage that it reduces the fault MVA level of the power sys.

As compare to a TF of low voltage rating, a leakage reactance of a high voltage rated TF is greater because its thicker insulation makes the wdggs further apart.



(attractive)



(Repulsive)

EFFICIENCY

$$\eta = \frac{\text{Output}}{\text{Input}}$$

$$= \frac{\text{O/P}}{\text{Output} + \text{loss}} \quad (\text{OR}) \quad \frac{\text{Input} - \text{losses}}{\text{Input}}$$

$$\left(1 - \frac{\text{losses}}{\text{Input}}\right)$$

TF has 25% loss then η is 80%.

$$\eta = \frac{\text{O/P}}{\text{O/P} + 0.25 \text{O/P}}$$

$$= \frac{\text{O/P}}{1.25 \text{O/P}}$$

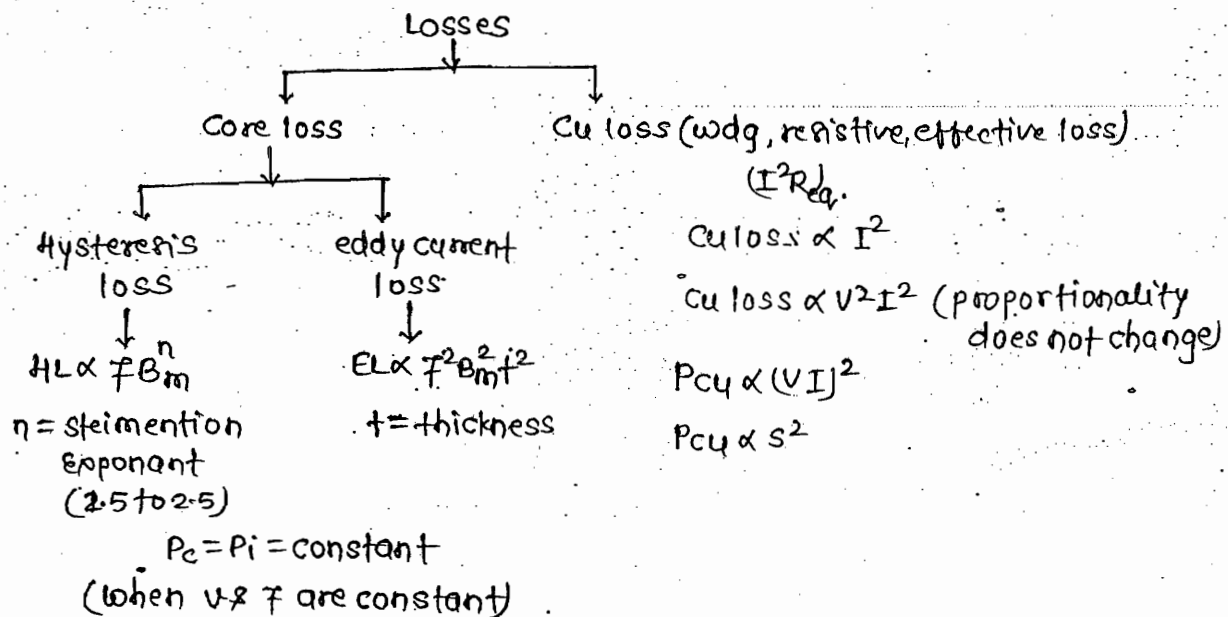
$$\boxed{\eta = 80\%}$$

100% loss then $\eta = 50\%$

$$\eta = \frac{\text{O/P}}{\text{O/P} + 1 \text{O/P}}$$

$$\eta = \frac{\text{O/P}}{2 \text{O/P}}$$

$$\boxed{\eta = 50\%}$$



$$\boxed{\eta = \frac{VI \cos \phi}{VI \cos \phi + I^2 R_{eq} + P_i}}$$

max^m efficiency →

Sagar Sen
887143536

$$\eta = \frac{V I \cos \phi}{V I \cos \phi + I^2 R_{eq} + P_i}$$

If the load current is constant but the load PF is variable then the max^m η of TF is obtained when the load PF is UNITY.

However if the load PF is constant but the load current is variable then the condⁿ for max^m η is obtained as follows:-

Dividing the numerator & denominator by 'I'

$$\eta = \frac{V \cos \phi}{V \cos \phi + I R_{eq} + \frac{P_i}{I}}$$

The η is max^m when the denominator is min^m.

$$\text{i.e. } \frac{d\eta(\text{denominator})}{dI} = 0$$

$$\frac{d}{dI} (V \cos \phi + I R_{eq} + \frac{P_i}{I}) = 0$$

$$0 + R_{eq} - \frac{P_i}{I^2} = 0$$

$$R_{eq} I^2 = P_i$$

$$\boxed{\text{Cu loss} = \text{Core loss}}$$

$$\boxed{I_{n\max} = \sqrt{\frac{P_i}{R_{eq}}}}$$

$$V I_{n\max} = V \sqrt{\frac{P_i}{R_{eq}}}$$

$$S_{n\max} = V \cdot I_j \sqrt{\frac{P_i}{I_j^2 R_{eq}}}$$

$$= S_j \sqrt{\frac{P_i}{P_{cu(j)}}}$$

S_j = Any kVA at which the cu loss is already available.

$$\boxed{S_{n\max} = S_j \sqrt{\frac{P_i}{P_{cu(j)}}}}$$

Example:-

(i) $s_j = s_{FL}$

$$\text{Then } S_{\eta_{max}} = S_{FL} \times \sqrt{\frac{P_i}{P_{cu}(FL)}}$$

(ii) $s_j = s_{(3\%)}$

$$\text{Then } S_{\eta_{max}} = s_{(3\%)} \times \sqrt{\frac{P_i}{P_{cu}(FL)}}$$

Alternative for $S_{\eta_{max}} \rightarrow$

$$P_{cu} \propto s^2$$

$$\frac{P_{cu}(\eta_{max})}{P_{cu}(j)} = \left(\frac{S_{\eta_{max}}}{s_j} \right)^2$$

$$\sqrt{\frac{P_i}{P_{cu}(j)}} = \frac{S_{\eta_{max}}}{s_j}$$

$$S_{\eta_{max}} = s_j \sqrt{\frac{P_i}{P_{cu}(j)}}$$

$$\eta_{max} = \frac{S_{\eta_{max}} \cos \phi}{S_{\eta_{max}} \cos \phi + 2P_i}$$

DATE-06/11/19

All Day efficiency

(OR)

Energy efficiency

$$\eta_{\text{all-day}} = \frac{\text{o/p kwh in 24 hrs}}{\text{i/p kwh in 24 hrs.}}$$

$$= \frac{\text{o/p kwh in 24 hrs.}}{\text{o/p kwh in 24 hrs.} + P_{cu}(\text{kwh}) \text{ in 24} + P_i(\text{kwh}) \text{ in 24 hrs.}}$$

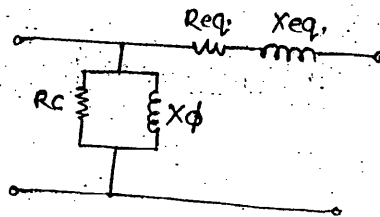
24 hrs.

(OR)

$$P_i(\text{kW}) \times 24$$

1.16(39)

Que. → for a 200kVA 4000/1000V 1-φ TF draw the eq. ckt referred to LV side & insert all the values.



It is given that the TF η at UPF is 97% both at FL & at 60% of FL. The NL PF is 0.25 & FL reg η at a lagging PF of 0.8 is 5%.

Soln →

$$0.97 = \frac{VI}{VI + P_c + P_i}$$

$$0.97 = \frac{(0.6)^2 VI}{(0.6)VI + P_c + P_i}$$

$$0.05 = \frac{V_i \eta - V}{V}$$

$$\frac{0.97}{200k} = \frac{1}{200k + P_c + P_i}$$

$$\frac{0.97}{0.6 \times 200 \times 10^3} = \frac{1}{0.6 \times 200 + 0.36 P_c + P_i}$$

$$P_c + P_i = 6.186 \times 10^3$$

$$0.36 P_c + P_i = 3.711 \times 10^3$$

$$P_c = 3865.625 \text{ W}$$

$$P_i = 2319.375 \text{ W}$$

$$0.97 = \frac{1 \times 1.0}{1 \times 1.0 + P_c(\text{FL}) + P_i}$$

$$P_c(\text{FL}) + P_i = 0.0309 \text{ PU} \quad \text{--- (i)}$$

$$0.97 = \frac{0.6 \times 1.0}{0.6 \times 1.0 + (0.6)^2 P_c(\text{FL}) + P_i}$$

$$0.36 P_c(\text{FL}) + P_i = 0.0186 \text{ PU} \quad \text{--- (ii)}$$

Solving eqn (1) & (2)

$$P_i = 0.0117 \text{ pu}$$

$$P_c(\text{pu}) = 0.0192 \text{ pu}$$

$$P_c(\text{pu}) = R_{\text{pu}} = 0.0192 \text{ pu}$$

$$R_{\text{pu}} = 0.0192 \text{ pu}$$

$$V_R^n = 0.05 \text{ pu} = R_{\text{pu}} \cos \phi + X_{\text{pu}} \sin \phi$$

$$0.05 = 0.0192 \times 0.8 + X_{\text{pu}} \times 0.6$$

$$X_{\text{pu}} = 0.0577 \text{ pu}$$

$$X_{\text{pu}} = 0.0577 \text{ pu}$$

$$P_i = \frac{V^2}{R_c(\text{pu})}$$

$$R_c(\text{pu}) = \frac{V^2}{P_i}$$

$$0.0117 = \frac{1^2}{R_c(\text{pu})}$$

$$R_c(\text{pu}) = 85.47 \text{ pu}$$

Given: $\cos \phi_0 = 0.25$

$$\phi_0 = 75.5224^\circ$$

$$X_{\phi}(\text{pu}) = \frac{R_c(\text{pu})}{\tan \phi_0}$$

$$\therefore \tan \phi = \frac{R}{X}$$

$$X_{\phi}(\text{pu}) = \frac{85.47}{\tan(75.52^\circ)} = 22.07 \text{ pu}$$

$$X_{\phi}(\text{pu}) = 22.07 \text{ pu}$$

$$Z_{\text{Base}}(\Omega) = \frac{(100)^2}{200 \times 10^3} = 5 \Omega$$

Actual Ω = pu value $\times$ Base impedance

$$X_{\phi}(\text{act}) = 22.07 \times 5 = 110.35 \Omega$$

$$R_c(\text{act}) = 85.47 \times 5 = 427.15 \Omega$$

$$R_{eq}(\text{act}) = 0.0577 \times 5 = 0.2885 \Omega$$

$$R_{eq}(\text{act}) = 0.0192 \times 5 = 0.096 \Omega$$

Que → The max^m η of 500kVA, 3300/500V 50Hz 1- ϕ TF is 97% & occurs at 75% FL UPF. If the TF impedance is 10%, calc. the regⁿ at 0.8 PF lagging.

Soln →

$$0.97 = \frac{P_i}{P_i + 2 \times P_c} = \frac{0.75 \times 1}{0.75 \times 1 + 2 \times P_c}$$

$$P_i = P_c = 0.0115 \text{ PU} = P_c(75\%)$$

$$I_{\eta \max} = \sqrt{\frac{P_i}{R_{eq}}}$$

$$S_{\eta \max} = 0.0115$$

$$\begin{aligned} P_{cu}(75\%) &= \frac{P_{cu}(75\%)}{(0.75)^2} \\ &= \frac{0.0116}{(0.75)^2} \\ &= 0.0206 \text{ PU} \end{aligned}$$

$$Z_{pu} = 0.1 \angle \cos^{-1} \frac{0.0206}{0.1}$$

$$Z_{pu} = 0.10 \angle 78.11^\circ \text{ PU}$$

$$R_{eq} = Z_{pu} \cos(\theta_{eq} - \phi)$$

$$= 0.10 \cos[78.11 - \cos^{-1}(0.8)]$$

$$R_{eq} = 0.0752 \text{ PU}$$

$$= 75.2\%$$

Que → A 500kVA TF has a max^m η of 98.6% at 350kVA UPF. During the day it is loaded as follows:-

- (1) 6hrs: 300kVA, 0.8 PF lag
- (2) 4hrs: 240kW, 0.6 PF lead
- (3) 5hrs: No load
- (4) 9hrs: 225kVA UPF

Calc. the all day η of TF.

Solⁿ→

$$0.986 = \frac{350 \times 1.0}{350 \times 1.0 + 2P_i}$$

$$P_i = 2.4848 \text{ kW} = P_{cu}(350)$$

6hrs:-

$$O/p \text{ kWh} = (300 \times 0.8) \times 6 \text{ hrs} = 1440 \text{ kWh}$$

$$P_{cu}(\text{kWh}) = \left[2.4848 \times \left(\frac{300}{350} \right)^2 \right] \times 6$$
$$= 10.953 \text{ kWh}$$

4hrs:-

$$O/p \text{ kWh} = 240 \times 4 = 960 \text{ kWh}$$

$$P_{cu}(\text{kWh}) = 2.4848 \times \left[\frac{(240/0.6)}{350} \right]^2 \times 4$$
$$= 12.082 \text{ kWh}$$

5hrs:-

$$O/p \text{ kWh} = 0; \quad P_{cu}(\text{kWh}) = 0$$

9hrs:-

$$O/p \text{ kWh} = (225 \times 1.0) \times 9 = 2025 \text{ kWh}$$

$$P_{cu}(\text{kWh}) = \left[2.4848 \times \left(\frac{225}{350} \right)^2 \right] \times 9$$
$$= 9.242 \text{ kWh}$$

Total during the day (24 hrs)

$$O/p \text{ kWh} = 442.5 \text{ kWh}$$

$$P_{cu}(\text{kWh}) = 33.177 \text{ kWh}$$

$$P_i(\text{kWh}) = 2.4848 \times 24 = 59.6352 \text{ kWh}$$

$$\eta_{\text{all-day}} = \frac{442.5}{442.5 + 33.177 + 59.6352}$$

$$= 0.9795 \text{ pu}$$

$$\boxed{\eta_{\text{all-day}} = 97.9\%}$$